

Product Requirements Document

EduAI

Pakistan's First Agentic AI Learning Platform

Document Version: 1.0

Date: August 9, 2026

Status: Draft

Owner: Product Team

1. Executive Summary

EduAI is an AI-powered education platform built specifically for Pakistani students following the PTB (Punjab Textbook Board) / FBISE (Federal Board of Intermediate and Secondary Education) curriculum. It functions as a personal AI tutor — explaining concepts in Urdu and English, generating quizzes and notes, detecting weak topics, and building adaptive study plans. Teachers monitor classroom performance through dedicated dashboards, and parents receive real-time progress updates. The platform uses multiple cooperating AI agents (an "agentic" architecture) and supports offline learning, aiming to make quality, curriculum-aligned education accessible to students anywhere in Pakistan.

2. Problem Statement

- Access to quality, curriculum-aligned tutoring is limited outside major urban centers in Pakistan.
- Private tutoring is expensive and inconsistent in quality; class sizes leave little room for individualized attention.
- Students lack tools to identify their own weak topics and study efficiently ahead of board exams.
- Parents have little visibility into their child's actual academic progress between report cards.
- Existing ed-tech products are largely generic (US/India-curriculum focused) and not aligned to PTB/FBISE syllabi, nor do they support bilingual (Urdu/English) instruction.
- Unreliable and costly internet access in many regions makes always-online platforms impractical.

3. Goals & Objectives

Goal	Description
Curriculum alignment	100% content mapped to PTB/FBISE syllabi, subject-wise and grade-wise
Personalization	Every student gets an adaptive study plan based on their own performance data
Bilingual accessibility	All AI explanations available in Urdu and English
Multi-stakeholder visibility	Students, teachers, parents, and admins each get a role-specific experience
Offline-first	Core learning content usable without a continuous internet connection
Engagement	Increase time-on-task and topic mastery through quizzes, flashcards, and mind maps

Success Metrics (KPIs)

- Weekly Active Students (WAS) and average session duration
- % improvement in quiz scores per student over a grading period
- Weak-topic-to-mastery conversion rate (topics moved from "weak" to "mastered")
- Teacher dashboard weekly adoption rate
- Parent app open rate per week

- Offline content usage as % of total usage
- Student/parent Net Promoter Score (NPS)
- Churn rate (monthly/term-over-term)

4. Target Users & Personas

4.1 Student (Primary User)

Grades 6–12 (Matric/Intermediate), studying under PTB or FBISE boards. Needs help understanding concepts, revising before exams, and tracking personal progress. Comfortable with mobile phones; connectivity may be inconsistent.

4.2 Teacher

Manages one or more classes/sections. Needs to assign homework, view class-wide performance, identify struggling students, and reduce manual grading effort.

4.3 Parent/Guardian

Wants simple, real-time visibility into their child's attendance, quiz scores, weak areas, and homework completion — without needing to understand the platform in depth.

4.4 Admin (School/Platform Administrator)

Manages institutional accounts, user provisioning, content oversight, subscription/billing, and platform-wide analytics.

5. Scope

5.1 In Scope (v1.0)

All 20 modules listed in Section 6, for PTB and FBISE curricula, Grades 6–12, in Urdu and English.

5.2 Out of Scope (v1.0)

- Curricula outside PTB/FBISE (e.g., Cambridge/O-Level, other provincial boards) — future phase
- Live one-on-one human tutoring marketplace
- Payment/fee collection for schools (only subscription billing for the platform itself)
- Full native offline support for AI-generation features (only pre-downloaded/cached content works offline; generation requires connectivity)

6. Modules & Functional Requirements

6.1 Authentication (Student, Teacher, Parent, Admin)

- Role-based signup/login (email, phone/OTP, and school-issued credentials)
- Role-based access control (RBAC) with distinct permissions per role
- Parent-to-child account linking (via invite code or school roster)
- Teacher-to-class/section assignment by Admin
- Password reset, session management, multi-device login
- Optional single sign-on (SSO) for partner schools

6.2 Student Dashboard

- Overview of enrolled subjects, upcoming assignments, and today's study plan
- Quick access to AI Tutor, Quiz Generator, Notes, Flashcards
- Progress snapshot (streaks, mastery %, weak topics)
- Notifications feed

6.3 Teacher Dashboard

- Class/section overview with per-student performance heatmap
- Assignment creation, distribution, and auto-graded results
- Weak-topic reports aggregated at class level
- Attendance marking and history
- Ability to message/notify parents or students

6.4 Parent Dashboard

- Real-time view of child's attendance, quiz results, and homework status
- Weekly/monthly progress summary (auto-generated, plain-language)
- Weak topic alerts and suggested actions
- Direct notifications for major events (missed homework, low quiz score, exam reminders)

6.5 Admin Panel

- User & role management (bulk import via CSV, school roster sync)
- Content management (curriculum mapping, subject/chapter structure)
- Subscription & billing management
- Platform-wide analytics (usage, engagement, retention)
- Audit logs and system health monitoring

6.6 AI Tutor

- Conversational, curriculum-aware tutor per subject/chapter
- Step-by-step concept explanations in Urdu, English, or mixed (Roman Urdu supported)
- Follows PTB/FBISE textbook sequencing and terminology
- Context-aware follow-up Q&A within a topic/session

6.7 AI Voice Assistant

- Voice-based interaction with the AI Tutor (speech-to-text and text-to-speech)
- Urdu and English voice support
- Hands-free study mode for revision/listening-based learning

6.8 AI Study Planner

- Generates personalized, adaptive daily/weekly study schedules
- Adjusts plan based on quiz performance, exam dates, and weak topics
- Reminders integrated with Notifications module
- Supports goal-based planning (e.g., "prepare for Physics mid-term in 3 weeks")

6.9 Smart Quiz Generator

- Auto-generates quizzes by subject, chapter, topic, and difficulty level
- MCQs, short-answer, and long-answer question types aligned to board exam patterns
- Instant grading and explanation of correct answers
- Difficulty adapts based on student's historical performance

6.10 Weak Topic Detection

- Analyzes quiz/assignment results to flag topics below a mastery threshold
- Feeds directly into the Study Planner and Teacher/Parent dashboards
- Visual weak-topic map per subject

6.11 AI Notes Generator

- Auto-generates concise, curriculum-aligned notes per chapter/topic
- Available in Urdu and English
- Exportable/downloadable for offline use

6.12 Flashcards & Mind Maps

- AI-generated flashcards for key terms, formulas, and definitions
- Auto-generated mind maps to visualize chapter/topic relationships
- Spaced-repetition review mode for flashcards

6.13 PTB/FBISE Digital Library

- Digitized textbooks and reference material aligned to official PTB/FBISE syllabi
- Searchable by grade, subject, and chapter
- Downloadable for offline access

6.14 Assignment & Homework Management

- Teachers create and assign homework with due dates
- Students submit work (text, file upload, or auto-graded quiz format)
- Auto-grading for objective assignments; manual grading workflow for subjective ones

- Status tracking (assigned, submitted, graded, overdue)

6.15 Progress & Performance Analytics

- Student-level, class-level, and school-level analytics
- Trend charts for scores, attendance, and engagement over time
- Exportable reports for teachers/admins

6.16 Attendance Management

- Daily attendance marking by teachers (manual or QR/code-based)
- Attendance history and reports visible to teachers, parents, and admins
- Automated absence alerts to parents

6.17 AI Doubt Solver

- Students can ask specific questions (text, or photo of a textbook problem) and receive step-by-step solutions
- Curriculum-aware, shows working/steps rather than only the final answer
- Escalation path to a human teacher if the AI cannot resolve the doubt confidently

6.18 Offline Learning

- Downloadable content packs (notes, flashcards, library material, pre-generated quizzes) per subject/chapter
- Offline progress syncs automatically once connectivity is restored
- Low-bandwidth mode for constrained connections

6.19 Notifications & Reminders

- Push/SMS/email notifications for study plan tasks, homework deadlines, exam reminders, attendance alerts, and progress summaries
- Configurable notification preferences per role

6.20 Profile & Settings

- Editable profile info per role
- Language preference (Urdu/English/Roman Urdu)
- Notification preferences, privacy settings, account/data management
- Accessibility settings (font size, voice speed, etc.)

7. Agentic AI Architecture (High-Level)

EduAI is designed around cooperating specialized AI agents rather than a single monolithic model call:

Agent	Responsibility
Tutor Agent	Conceptual explanations, conversational teaching
Quiz Agent	Question generation, grading, difficulty calibration
Planner Agent	Adaptive scheduling based on performance data
Diagnostic Agent	Weak topic detection and mastery scoring
Notes/Flashcard Agent	Content summarization and study-material generation
Voice Agent	Speech-to-text / text-to-speech orchestration
Doubt-Solving Agent	Step-by-step problem solving, including image-based input

These agents share a common student performance/context store so that, for example, a weak topic detected by the Diagnostic Agent immediately informs the Planner Agent's next schedule and the Quiz Agent's next question difficulty.

8. Non-Functional Requirements

- Localization: Full Urdu and English UI and content support; Roman Urdu input handling for AI Tutor/Doubt Solver.
- Performance: AI responses returned within a few seconds under normal connectivity; graceful degradation on low bandwidth.
- Offline capability: Core content (library, notes, flashcards, downloaded quizzes) accessible without internet; sync on reconnect.
- Scalability: Architecture must support scaling from individual students to full school/district rollouts.
- Security & Privacy: Student data (especially minors' data) protected with encryption at rest/in transit, RBAC, and compliance with applicable data protection norms; parental consent flows for minors.
- Reliability: Target uptime of 99.5%+ for core services.
- Device support: Android-first (given market share in Pakistan), with responsive web support; low-end device optimization.
- Accessibility: Adjustable text size, voice-assist for low-literacy users.

9. Assumptions & Dependencies

- Access to digitized/licensed PTB and FBISE curriculum content.
- Reliable Urdu-language speech-to-text/text-to-speech capability (via third-party or custom model).
- Partnering schools willing to provide rosters for bulk onboarding.
- Underlying AI model provider(s) selected separately (not specified in this PRD).

10. Risks

Risk	Impact	Mitigation
Low internet penetration in target regions	Reduced engagement	Prioritize offline-first design
Curriculum content licensing delays	Delayed launch	Start content partnerships early
Urdu AI quality (voice/text) below expectations	Poor user experience	Dedicated QA/testing for Urdu language features before launch
Low digital literacy among parents	Low parent-dashboard adoption	Simple UI, SMS-based summaries as fallback
Data privacy concerns (minors)	Trust/legal risk	Strong consent flows, minimal data collection, encryption

11. Release Phasing (Suggested)

Phase 1 (MVP)

Authentication, Student Dashboard, AI Tutor, Smart Quiz Generator, AI Notes Generator, Progress Analytics, Notifications

Phase 2

Teacher Dashboard, Assignment & Homework Management, Weak Topic Detection, AI Study Planner, Attendance Management

Phase 3

Parent Dashboard, Flashcards & Mind Maps, AI Voice Assistant, AI Doubt Solver, PTB/FBISE Digital Library

Phase 4

Offline Learning (full rollout), Admin Panel (full analytics suite), scaling to additional boards/curricula

12. Open Questions

- Which grades/subjects launch first — Matric (9-10) or Intermediate (11-12), or both simultaneously?
- Will school partnerships be B2B (institutional licenses) or B2C (individual student subscriptions), or both?
- What is the monetization model — freemium, subscription tiers, or school-license only?
- Which AI/voice providers will be used for Urdu speech support?

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